

NETWORK PROBLEM 3:

MAC FLOODING (LAYER 2 DATA LINK LAYER)

MAC Flooding is a type of network attack where a switch's MAC address table is overwhelmed with fake MAC addresses, causing the switch to behave like a hub. This leads to serious security and performance issues in the network.

Causes of MAC Flooding:

1. Switch Table Limitations:

- Switches maintain a MAC address table (CAM table) that maps MAC addresses to specific ports. This table has a finite size.
- An attacker sends frames with different source MAC addresses rapidly, exhausting the table's capacity.

2. Malicious Attack Tools:

- Tools like macof or Yersinia can generate thousands of fake MAC addresses per second.
- These tools are used by attackers to intentionally flood the switch.

3. Lack of Port Security:

- If port security is not enabled on switch ports, the switch will learn and store unlimited MAC addresses from any source.
- This allows attackers to exploit the switch freely.

4. Poor Network Segmentation:

- Networks that are flat and lack VLAN separation are more vulnerable to such broadcast-based attacks.

Symptoms of MAC Flooding:

1. Broadcast Behavior:

- Once the MAC table is full, the switch starts broadcasting all incoming traffic to all ports.
- This compromises data privacy and allows attackers to sniff packets not intended for them.

2. Network Performance Degradation:

- With excessive traffic being sent to all ports, network congestion increases. o
Devices experience latency, dropped packets, or even total communication failure.

3. Unauthorized Access to Data:

- Attackers can intercept sensitive data like login credentials or personal information from broadcast traffic.

4. Unusual Traffic Patterns:

- Network monitoring tools may detect a high volume of unknown MAC addresses and unexpected traffic on multiple switch ports.

Resolving MAC Flooding:

1. Enable Port Security:

- Configure port security on switches to limit the number of MAC addresses learned per port.
- Example: Limit each port to 1–2 allowed MAC addresses.

2. Use Static MAC Address Binding:

- Bind known MAC addresses to specific switch ports. This prevents unknown devices from communicating through protected ports.

3. Implement VLANs:

- Use Virtual LANs to logically segment the network.
- This minimizes the attack surface by isolating devices.

4. MAC Address Table Monitoring:

- Regularly monitor the MAC address table for anomalies. Suspiciously high numbers of learned MAC addresses should trigger alerts.

5. Use Managed Switches with Security Features:

- Deploy switches that support features like Dynamic ARP Inspection (DAI), DHCP Snooping, and port-based authentication (802.1X).

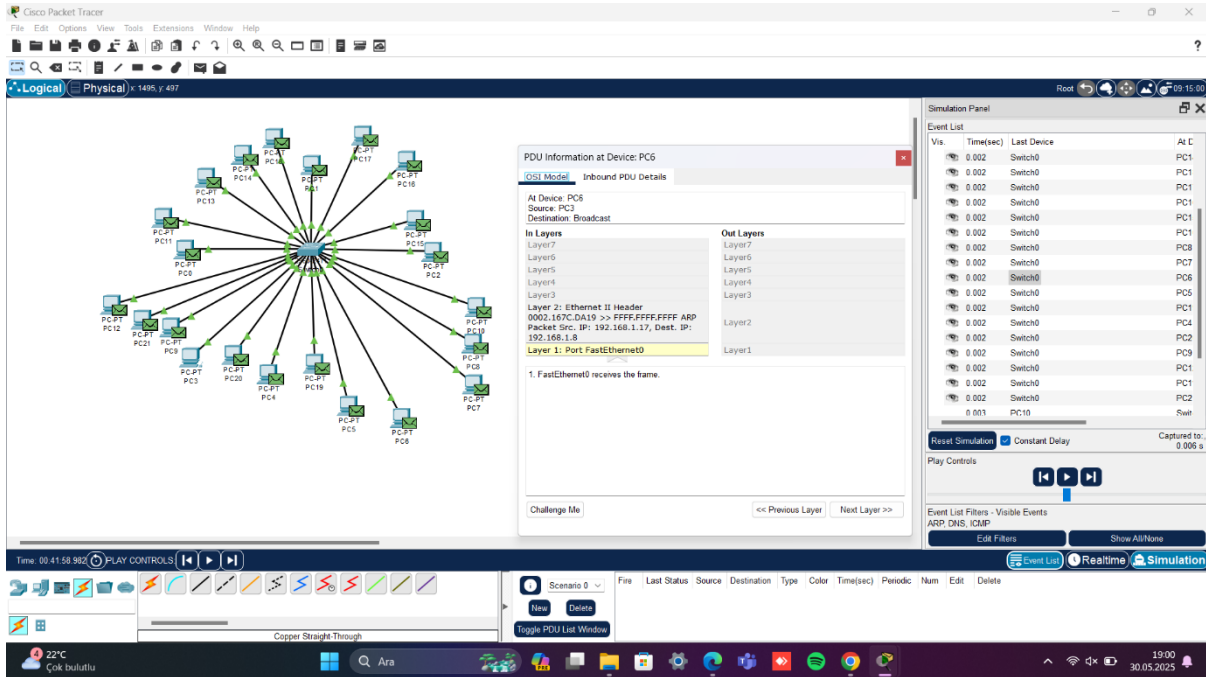
6. Intrusion Detection and Prevention Systems (IDS/IPS):

- Use IDS/IPS to detect and block MAC flooding attacks in real-time.

7. Educate Network Users:

- Awareness training for administrators and users helps reduce insider threats and misuse of network tools.

Simulation of problem: In this simulation, 21 PCs were used to send pings to a target PC from 20 PCs. As a result, broadcast behavior, which is one of the symptoms of Mac flooding, was observed.



Simulation of solution: Using first solution(Enable Port Security)